

Density-dependent habitat selection of spawning Chinook salmon: broad-scale evidence and implications

Matthew R. Falcu*

Oregon Department of Fish and Wildlife, 4034 Fairview Industrial Dr SE, Salem, OR 97302, USA

Summary

1. An extensive body of theory suggests that density-dependent habitat selection drives many fundamental ecological processes. The ideal free distribution and the ideal despotic distribution make contrasting predictions about the effect of total population size on relative abundances among habitats. Empirical assessment of these habitat selection models is uncommon because data must be collected over large temporal and spatial scales.

2. I ask whether fluctuation in Chinook salmon (*Oncorhynchus tshawytscha*) spawner population size through time leads to different relative densities over space.

3. Twenty-six years of monitoring data on spawning Chinook salmon across the entire coast of Oregon, USA, were used to evaluate models that make contrasting statements about the interactions of a latent population abundance parameter with physical habitat characteristics.

4. There is strong information-theoretic support for models that include terms that allow the spatial variation in density to change as population size changes through time. Analysis of the best model reveals nonlinear isodars, which suggests a ‘despotic’ or ‘preemptive’ distribution of individuals across habitats, indicating that dominant or early-arriving individuals exclude others from breeding sites.

5. This finding has implications for genetic dynamics, population dynamics and conservation metrics of these highly valued fish. The novel application of modelling techniques used here to assess mechanisms of habitat selection from observational data can be used in the emerging field of eco-evolutionary dynamics.

Key-words: habitat selection, density dependence, ideal free distribution, despotic distribution, isodar, Chinook salmon, hierarchical model

Introduction

Mechanisms of habitat selection can influence basic ecological processes such as population regulation (Pulliam 1988, Pulliam & Danielson 1991; Rodenhouse, Sherry & Holmes 1997; McPeck *et al.* 2001) and species interactions (Rosenzweig 1991; Danielson 1992). These ecological processes, in turn, affect evolutionary processes such as natural selection, genetic drift and gene flow (Holt 1987). If rapid evolution has a significant effect on ecological processes (Thompson 1998; Schoener 2011), then habitat selection can have a fundamental role in the resulting feedback loop between ecology and evolution (Holt 1987; Morris 2003, 2011). Mechanisms of habitat selection are therefore important to the emerging field of eco-evolutionary dynamics.

The ideal free distribution (IFD; Fretwell & Lucas 1970) arises when awareness of all habitat qualities is ‘ideal’ (perfect) and animals are ‘free’ to move among habitats without reducing fitness. The IFD results in equal fitness among individuals in different habitats because any inequality would be offset by dispersal and negative density effects. The IFD assumes that animals do not interfere with one another; however, if some animals pre-emptively select the best breeding sites and then successfully defend them or, more generally, there are dominant animals that exclude subordinates from the best breeding sites (ideal despotic distribution; IDD; Fretwell & Lucas 1970), then the quality of the poorest available breeding site will be the same in different habitats. Unlike the IFD, the IDD does not lead to equal fitness among individuals. Importantly, the territorial behaviour associated with the IDD results in different predictions than the IFD about the effect of population size on the distribution of animals among habitats.

*Correspondence author. E-mail: matthew.r.falcu@state.or.us

The IFD leads to a linear relationship between densities of individuals in two habitats as total population size varies (an 'isodar'; Morris 1998; Fig. 1a,b), whereas the IDD results in curved isodars (Fig. 1c–h). Thus, only the IDD predicts change in the relative densities of animals among habitats as total population size varies. If spatial variation in animal density is conceived as peaks and valleys of a topographic map, the IFD predicts that increase in total population size results in a scalar increase in elevation across the entire map. The IDD predicts that increase in total population size also causes the locations of peaks and valleys to shift in space. It should therefore be possible to detect whether animals follow an IFD or IDD from observations of local abundance as total population size varies over time. Despite the importance of mechanisms of habitat selection, determining whether total population size affects the distribution of individuals among habitats is uncommon.

Chinook salmon (*Oncorhynchus tshawytscha*) are well suited to the analysis of density-dependent habitat selection. Chinook salmon are anadromous and semelparous; thus, every year a population of salmon will return from the ocean to freshwater, select sites for spawning and then die. Chinook salmon also exhibit large interannual variation in spawning (adult) population size. These life-history characteristics are well suited to empirical assessment of habitat selection because the spawning component of populations displays temporally discrete habitat selection events every year, with high contrast in abundance across years.

Here, 2436 observations of Chinook salmon density made over 26 years and across 14 populations are used to evaluate multiple models that make contrasting statements

about how these fish use space. Spatial variation in the intrinsic quality of spawning habitat should be a function of geomorphology (Montgomery *et al.* 1999; Burnett *et al.* 2007; Busch *et al.* 2013) as well as the volume of water flowing through the catchment system during a spawning run (Hodgson *et al.* 2006). Upon this template of spatiotemporal variation of spawning habitat quality, I assessed the effect of population size on the distribution of Chinook salmon among spawning habitats.

Materials and methods

The Oregon Department of Fish and Wildlife conducts annual Chinook salmon spawning surveys at numerous sites along the Oregon Coast. At approximately weekly intervals during the fall spawning run, living and dead fish within a given site (typically a 1.6-km stream segment) are visually enumerated. The maximum of these repeat counts within a site and year is known as the 'peak count', which is the response variable modelled here.

Most of the sites were chosen based on ease of access and reliability of detecting fish. In 1986, the number of sites where peak counts were recorded sharply increased. Thus, the peak count data analysed here begin in 1986 and continue through 2011. Peak counts recorded at eight index surveys within one population are given in Fig. 2. In addition to index sites with relatively continuous time series, peak counts of spawning Chinook salmon were also recorded at sites selected with a spatially randomized survey design. The sample frame for drawing spatially randomized survey sites was designed for coho salmon (*Oncorhynchus kisutch*; ODFW 2002), which extends farther upstream than the spatial extent of spawning Chinook salmon. In addition, peak counts of Chinook salmon have been recorded in river mainstems. During the 26-year time period analysed here, 2436

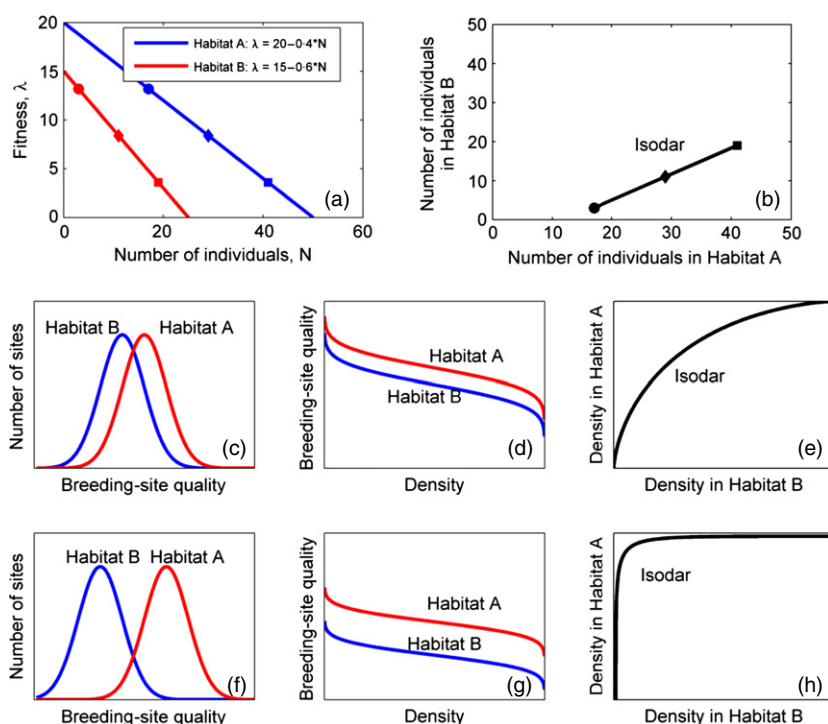


Fig. 1. Animals obeying an ideal free distribution should have identical fitnesses in two habitats. As total population size changes from 20 to 40 and to 60 (circle, diamond, square, respectively), individuals' fitness declines but remains identical across habitats (a). A plot of the number of individuals in each habitat across different population sizes is known as an isodar (b). Isodars are linear as long as the lines in (a) are not curved. The ideal pre-emptive habitat selection theory postulates that newly arriving individuals will occupy the best vacant site (move right to left in c and f). The negative inverse cumulative distribution (d and g) of the distributions in (c) and (f) is the fitness of individuals in the two habitats as population size increases. The isodars that represent densities of individuals in these two habitats are curved (e and h). Greater contrast in mean site quality of two habitats (c vs. f) leads to greater nonlinearity in the resulting isodar (e vs. h). Adapted from Morris (1994).

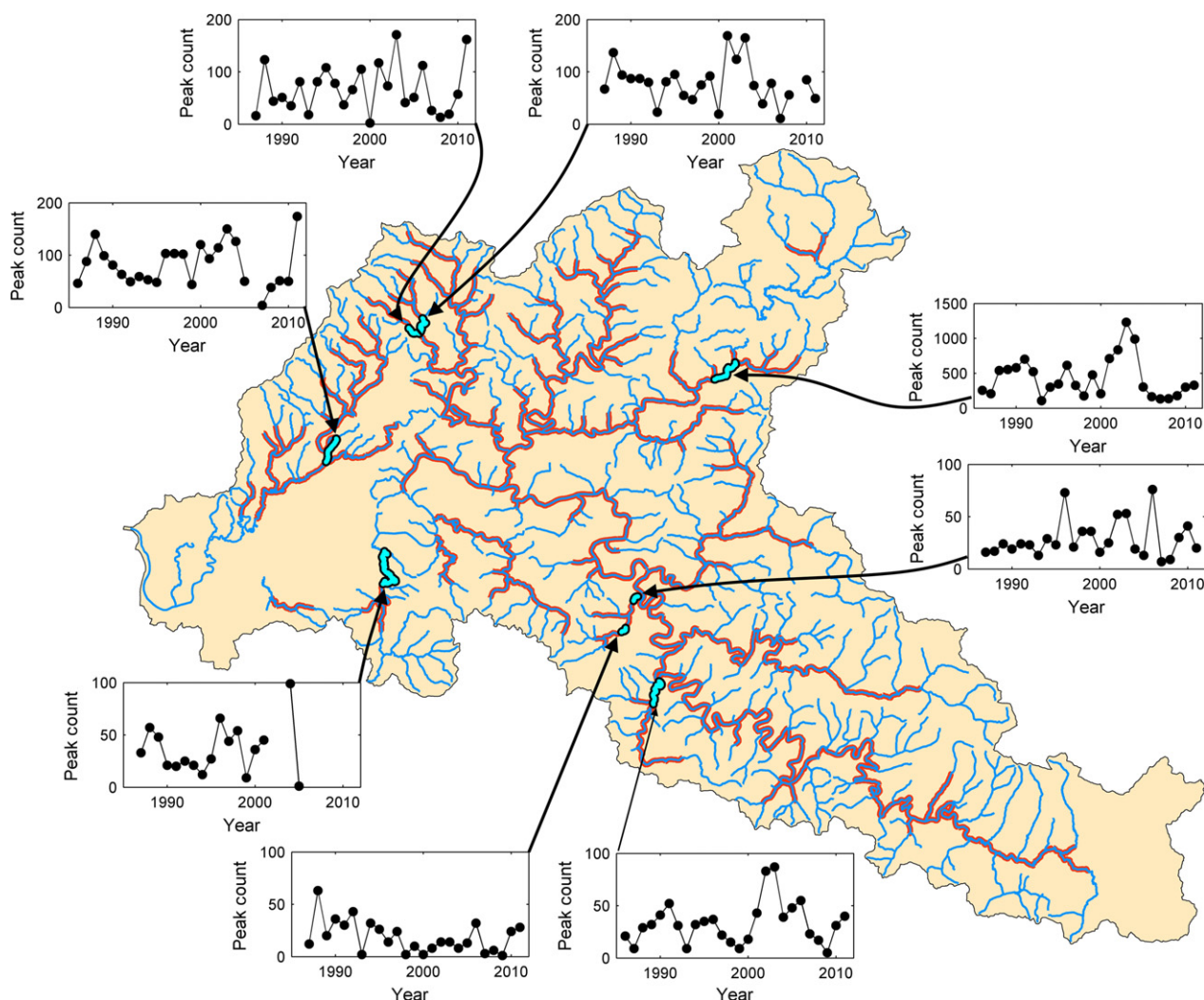


Fig. 2. Peak count time series recorded at eight fixed index sites within a single Chinook population. Additional surveys that did not result in a time series were also used in the analysis, but are not shown here. Red/bold lines denote stream reaches thought to contain spawning habitat.

observations of peak count were made at 670 sites distributed among 14 populations. Population boundaries were defined by catchment systems with a common point of ocean entrance (Fig. S1, Supporting information).

Previous studies (Burnett *et al.* 2007; Busch *et al.* 2013) used literature review to infer the shape of habitat suitability functions over three geomorphic attributes: channel gradient ('gradient', %), valley width ('width', m) and mean annual stream flow ('MAF', m³/s). The same geomorphic attributes are used herein to fit models of observed peak counts of spawning Chinook. The geomorphic attributes were derived from 10-m digital elevation models (DEMs) and exist in a geographic information system (GIS) layer created by the Coastal Landscape Analysis and Modeling Study (<http://www.fsl.orst.edu/clams/index.html>).

The US Geological Survey has measured daily stream flow in seven of the 14 populations since 1986. The means of November measurements were used as predictors of peak counts in the nine most northern populations. The mean of November and December stream flow measurements was used as a predictor for the remaining five populations in the south. The decision to use a

later time period of stream flow to predict peak counts in more southern populations was made prior to analysis based on an observed latitudinal cline in the migration timing of Chinook salmon. If stream flow was not measured within a population, then the stream flow recorded in a nearby population was used as a surrogate (Fig. S1). The mean Pearson correlation coefficient of stream flows between nearby populations is 0.92. The geomorphic and stream flow covariates were standardized to zero mean, unit variance prior to model fitting.

Peak counts, y , at site i , year t , and population p were modelled with a Poisson distribution

$$y_{i,t,p} \sim \text{Poisson}(\lambda_{i,t,p}).$$

The log of the Poisson rate parameter, λ , was modelled as a function of survey length, $\mathcal{L}$, a normal random effect that is the same for all sites within a population on a given year, $\alpha_{t,p}$, and an intercept common to all sites and times, β_0 . Five models with different combinations of predictor variables were fitted to the peak count data (Table 1). Model #1 is as follows:

Table 1. Five candidate models (columns) of Chinook salmon density that include different predictor terms (rows). Shaded cells indicate that a given term was not included in the model. $\alpha_{t,p}$ is a random effect unique to all observations of density within a year and population that is interpreted as a metric of latent population size. Smaller Deviance Information Criteria (DIC) indicate better models. Point estimates and 95% highest probability density intervals are given for the best model

	Model				
Term	1	2	3	4	5
Length offset					
$\alpha_{t,p}$					
Intercept	3.75 (3.67, 3.87)				
MAF	0.80 (0.78, 0.82)				
MAF ²	-0.21 (-0.22, -0.20)				
Gradient	-0.06 (-0.10, -0.03)				
Gradient ²	-0.22 (-0.23, -0.20)				
Width	0.48 (0.46, 0.50)				
Width ²	-0.06 (-0.06, -0.05)				
Flow	0.05 (-0.04, 0.15)				
Flow*MAF					
Flow*Gradient	0.10 (0.07, 0.13)				
Flow*Width	-0.06 (-0.08, -0.04)				
$\alpha_{t,p}$ *MAF					
$\alpha_{t,p}$ *Gradient	0.30 (0.29, 0.32)				
$\alpha_{t,p}$ *Width	0.18 (0.16, 0.20)				
DIC =	91 542	92 441	93 939	106 328	106 328

$$\log(\lambda_{i,t,p}) = \log(\mathcal{L}_{i,t,p}) + \alpha_{t,p} + \beta_0 + \dots \beta_1 * MAF_{i,p} + \beta_2 * MAF_{i,p}^2 + \dots \beta_3 * Gradient_{i,p} + \beta_4 * Gradient_{i,p}^2 + \beta_5 * Width_{i,p} + \beta_6 * Width_{i,p}^2 + \dots \beta_7 * Flow_{i,p} + \beta_8 * Flow_{i,p} * Width_{i,p} + \beta_9 * Flow_{i,p} * Gradient_{i,p} + \dots \beta_{10} * \alpha_{t,p} * Gradient_{i,p} + \beta_{11} * \alpha_{t,p} * Width_{i,p}$$

eqn 1

where

$$\alpha_{t,p} \sim \text{Normal}(0, \sigma^2).$$

Note that there is no coefficient for survey length; it is used here as an offset. As noted with subscripts, the geomorphic attributes vary over space, but are fixed in time. Stream flow changes through time and across populations, but is the same for all sites within a population.

The random effect, $\alpha_{t,p}$, captures the effects of unmeasured deviations that vary across time and across populations. Thus, $\alpha_{t,p}$ can be regarded as a latent variable for population size, which is unique to each of the populations and varies through time. The variance of the random effect, σ^2 , reflects extra-Poisson variation in the peak count data.

Bayesian techniques were used to fit models. Using non-informative normal prior distributions on parameters β and a diffuse uniform prior on σ , the joint posterior probability distribution can be written as follows:

$$p(\beta, \lambda, \sigma^2 | \mathbf{X}, \mathbf{y}, \mathbf{m}_\beta, \mathbf{v}_\beta, l_\sigma, u_\sigma) \propto \prod_{j=1}^{2436} \text{Pois}(y_j | \lambda_j) \prod_{j=1}^{2436} \text{Norm}(\ln \lambda_j | \mathbf{X}, \beta, \sigma^2) \dots \times \text{Norm}(\beta | \mathbf{m}_\beta, \mathbf{v}_\beta) \text{Unif}(\sigma, l_\sigma, u_\sigma).$$

A priori neutrality in the value of coefficients, β , was specified with diffuse normal distributions with mean zero and standard

deviations arising from the uniform distribution on the interval [0, 6] (Gelman 2006). A uniform distribution on the interval [0, 6] was also used as a prior for the standard deviation of the random effect, α . The model was fitted with BUGS software (Bayesian inference using Gibbs sampling, Gilks, Thomas & Spiegelhalter 1994). For each model, three Markov chain Monte Carlo (MCMC) simulations were given a 1000 iteration 'burn-in'. Samples were 'thinned' to every 47th iteration, and total of 3000 samples per chain were obtained. Time series plots of MCMC samples were inspected visually for signs of convergence and mixing. Gelman–Rubin convergence diagnostics for all parameters were good ($\hat{R} = 1 \pm 0.0005$).

Only models 1–3 contain terms for an interaction between the random effect, $\alpha_{t,p}$, and geomorphic attributes (Table 1). Thus, only models 1–3 stipulate that population size ($\alpha_{t,p}$) affects the spatial distribution of spawner density. Models 4 and 5 do not contain the interaction between $\alpha_{t,p}$ and geomorphology and therefore do not entertain the possibility that spatial variation of spawner density changes with total population size. Discrimination among these two groups of models is a test of the hypothesis that the spatial distribution of densities depends on total population size. Different models were created within these two groups to help assess the predictive performance of model complexity. Model 3 is a reduced version of model 1. Model 4 contains just 1 less term than model 3, but does not contain a focal interaction with $\alpha_{t,p}$. Model 5 is an even further simplification of model 4. These models were created a priori without strong expectations of the relative predictive ability of the various terms. Across all models, gradient was used more often than the other geomorphic attributes because it is probably easier to understand how other, unmeasured habitat features covary with gradient than the other two geomorphic variables. The number of models fitted to the data was limited to 5 as a precaution against spurious results.

Plots of isodars typically consider two discretely different habitat types. Here, habitats vary continuously within a given geomorphic attribute, so habitat pairs were defined as positive and negative standard deviations away from mean channel gradient

and valley width. The best model, as determined by Deviance Information Criterion (DIC; Spiegelhalter *et al.* 2002), was then used to estimate density in these habitats pairs. Varying the random effect, α , simulates different population sizes, and changing the multiplier of standard deviations changes the magnitude of the difference between the habitat pairs. Additionally, a mathematical analysis of the parameter estimates of the best model was performed to explore the effects of changing stream flow and population size ($\alpha_{i,p}$) on the geomorphic habitat value where maximum density of spawners occurs. The mathematical analysis is a description of the best model and is therefore presented in the Results.

Results

There is a strong information-theoretic support for more complex models of observed peak counts. The models with the two lowest DIC scores contain quadratic effects of geomorphic attributes, interactions between stream flow and geomorphic attributes, as well as interactions between geomorphic attributes and random effect (Table 1). The coefficient of determination of the model with the lowest DIC (the best model) is $R^2 = 0.56$ (Fig. 3). Posterior probability distributions of coefficients of the best model indicate that all terms except stream flow are statistically significant, although stream flow is involved in significant interactions with geomorphic attributes (Table 1). The estimated coefficients for predictors modelled with quadratic effects produce concave down functions of peak count within the observed range of the predictors (Fig. 4). Increasing stream flow during the spawning run results in a shift of the maximal peak count towards habitats that are in steeper and in narrower valleys (Fig. 4 top row), which generally corresponds to locations farther upstream.

To obtain an expression for where maximal peak count occurs over gradient, one can take the partial derivative of the best model with respect to gradient

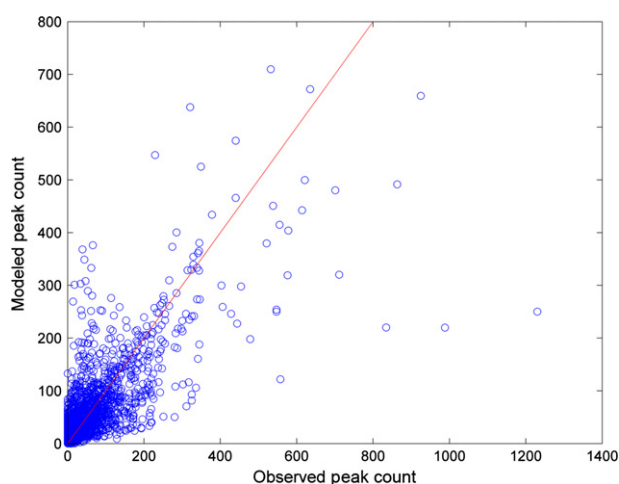


Fig. 3. Observed and predicted peak counts. Straight line indicates equivalence. Predicted peak counts made from the best model.

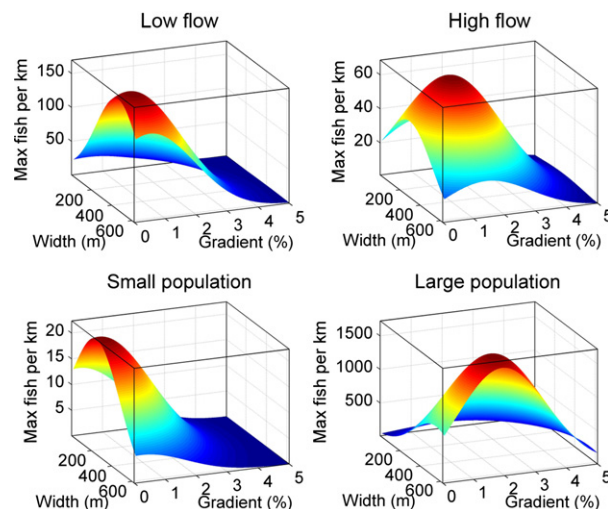


Fig. 4. The best model's predicted effects of two geomorphic variables (channel width and channel gradient) on fish density under different annual stream flow conditions (top row) and population sizes (bottom row).

$$\begin{aligned} \frac{\partial y}{\partial x_2} (y = & offset + \alpha + \beta_0 + \beta_1 x_1 + \beta_2 x_1^2 + \beta_3 x_2 \\ & + \beta_4 x_2^2 + \beta_5 x_3 + \beta_6 x_3^2 + \beta_7 x_4 + \beta_8 x_4 x_2 \\ & + \beta_9 x_4 x_3 + \beta_{10} \alpha x_2 + \beta_{11} \alpha x_3) \end{aligned}$$

where, for simplicity, terms in equation 1 are represented by x_i . The result is

$$\beta_3 + 2\beta_4 x_2 + \beta_8 x_4 + \beta_{10} \alpha.$$

Setting the expression above equal to zero and solving for the geomorphic variable yields

$$x_2 = \frac{-\beta_3 - \beta_8 x_4 - \beta_{10} \alpha}{2\beta_4}. \quad \text{eqn 2}$$

Similarly, the equation for the location of the maximal peak count over valley width, x_3 , is

$$x_3 = \frac{-\beta_5 - \beta_9 x_4 - \beta_{11} \alpha}{2\beta_6}. \quad \text{eqn 3}$$

Equations 2 and 3 indicate that the location of peaks of the surfaces in Fig. 4 over the geomorphic variables depends on stream flow, x_4 , and the latent indicator of population size, α . Maximum fish density was found at greater river gradients and lower widths as annual stream flow increased (Fig. 5a). Maximum fish density was found at greater river gradients and greater widths as population size increased (Fig. 5b).

Since the random effect is a latent indicator of population size, statistical interactions between the random effect and geomorphic attributes represent the effect of population size on spawning site selection. As population size

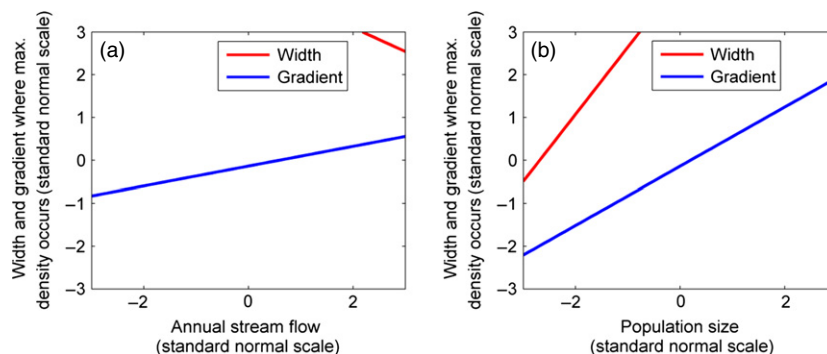


Fig. 5. The effect of annual stream flow (a) and population size (b) on the location of maximum fish density on physical habitat characteristics (valley width and channel gradient). Predictions made from analysis of best model.

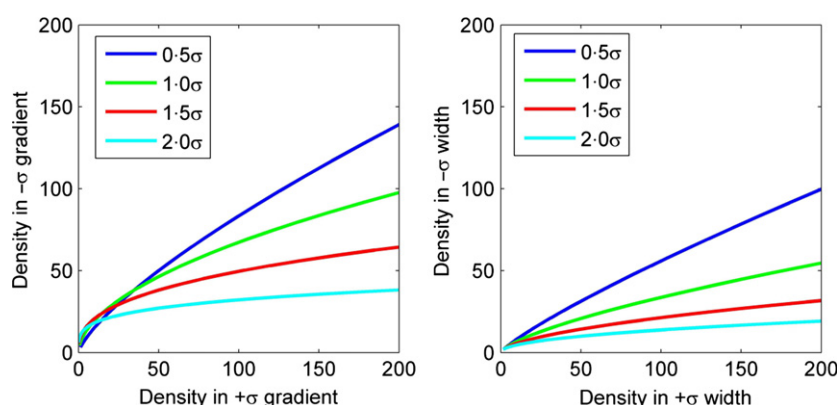


Fig. 6. Modelled isodars. Chinook density in two habitats (x and y axes) is plotted across a broad range of population size. Each panel presents a different habitat attribute (channel gradient on left, and channel width on right). Coloured lines give the contrast between the paired habitats. Contrast is measured by standard deviations, σ , on a standard normal scale. The blue/highest line on the left compares density in a habitat that is half a standard deviation below average gradient (y -axis) against density in a habitat that is half a standard deviation above average gradient (x -axis).

changes, so too does the spatial distribution of fish: Highest density sites occur in steeper streams and wider valleys as total population size increases (Fig. 4 bottom row, and Fig. 5b). The plot of paired densities (isodars) is nonlinear and sensitive to the contrast in habitat types (Fig. 6).

Discussion

To summarize, I used statistical methods to fit and rank competing models of observations of Chinook salmon density made over space and time. I then applied calculus to the best model to find the physical habitat attributes associated with maximum density as stream flow and total population size vary. The shape of isodars was assessed with the best model by plotting predicted abundances in pairs of habitats, which are nonlinear because of significant interaction terms included in the best model. In the remainder of the paper, I will discuss the following: (i) evidence of density-dependent habitat selection in related species, (ii) the effect of streamflow, (iii) inferential limitations of the modelling, (iv) interpretation of isodars, (v) the evolutionary and ecological significance of habitat selection, and (vi) management considerations.

To my knowledge, this is the first long-term and broad-scale examination of density-dependent habitat selection for adult Chinook salmon. Previous work on individual populations of juvenile Atlantic salmon has yielded support for an IFD (Hedger *et al.* 2005; Gibson, Bowlby & Amiro 2008), and Haugen *et al.* (2006) use a remarkable long-term data set to demonstrate that the relative abundances of pike (*Esox lucius*) in two habitats are consistent with empirically derived IFD predictions (see also Morris 2006). Still, Pacific salmon display life-history characteristics that should lead to an IDD. Females typically deposit all eggs into a single nest, or 'redd'. Mortality of developing embryos is relatively high and strongly related to redd site characteristics (broadly reviewed in Quinn 2005). Superimposition of one female's redd atop another's can cause significant density-dependent mortality (McNeil 1964; Fukushima, Quinn & Smoker 1998). Females should therefore experience strong selective pressure to use a high-quality redd site and defend it from others. Indeed, Pacific salmon are known to pre-emptively select preferred breeding sites (Dickerson *et al.* 2005), resist eviction (Foote 1990) and regularly attack one another on the spawning grounds (Quinn 1999; Esteve 2005).

Furthermore, females must deposit their eggs within a narrow temporal window set by the physiological processes of loosening eggs from the ovary. Females therefore cannot wait for residents of high-quality sites to senesce and die, so increasing density should lead to the use of progressively inferior sites (Quinn 2005, p. 110).

Stream flow conditions can affect the timing of a spawning run (Hodgson & Quinn 2002). Here, the best model's main effect for stream flow is not a significant predictor of spawning density because Chinook will spawn somewhere regardless of flow. The significant interaction of stream flow with the geomorphic attributes indicates that stream flow affects where Chinook spawn. The significant quadratic effects of valley width indicate that highest density sites occur at intermediate valley widths (Fig. 4). However, since most observations of Chinook density were made in narrow valleys, the highest density site occurs at a valley width that is underrepresented by these data. This can be seen in Fig. 5 where lines depicting maximum spawning density occur at extreme values of a standard normal distribution $|x| \gtrsim 3$.

The models I constructed are quantitative descriptions of how the spatial distribution of peak counts responds to total population size. Although these data cannot definitively illuminate specific mechanisms of habitat selection, the results are consistent with an IDD. Since the data are peak counts, it is not possible to address within-year space–time interactions. For example, peak counts within a given population and year are not necessarily synchronized, so some fish could hold in suboptimal habitats and then spawn in better habitat once density there decreases.

However, as noted above, physiological processes will restrict the extent to which this can occur.

Individual fish detection probability could not be assessed with these data. However, the only insignificant term in the best model presented here is stream flow, which would have been significant if detection probability strongly covaried with flow. Indeed, the Oregon Department of Fish and Wildlife estimates that stream visibility, stream depth, stream width and fish density have statistically insignificant effects on Chinook detection probability, which is 0.76 (ODFW 1984).

If animals within a habitat do not interfere with one another, as in the IFD, then the addition of individuals has an additive effect on all individuals' fitness in that habitat. When this occurs, isodars will be linear (Fig. 1a, b). Alternatively, if animals are territorial and interfere with one another, as in the IDD, then the quality of the poorest available breeding site will be the same in different habitats. The statistical distribution of breeding site qualities within habitats will result in a sigmoidal plot of mean fitness over population size, and the resulting isodar will be nonlinear (Morris 1994). The magnitude of curvature in the isodar will increase with increasing contrast in mean breeding site quality of two habitats (Fig. 1c–h). The isodars of spawning Chinook salmon computed here are moderately curvilinear in channel gradient, but nearly linear in valley width (Fig. 6). There are two non-exclusive explanations for the different curvatures in different kinds of habitats. First, the difference between mean breeding site qualities of two different channel-gradient habitats could be larger than that of two different

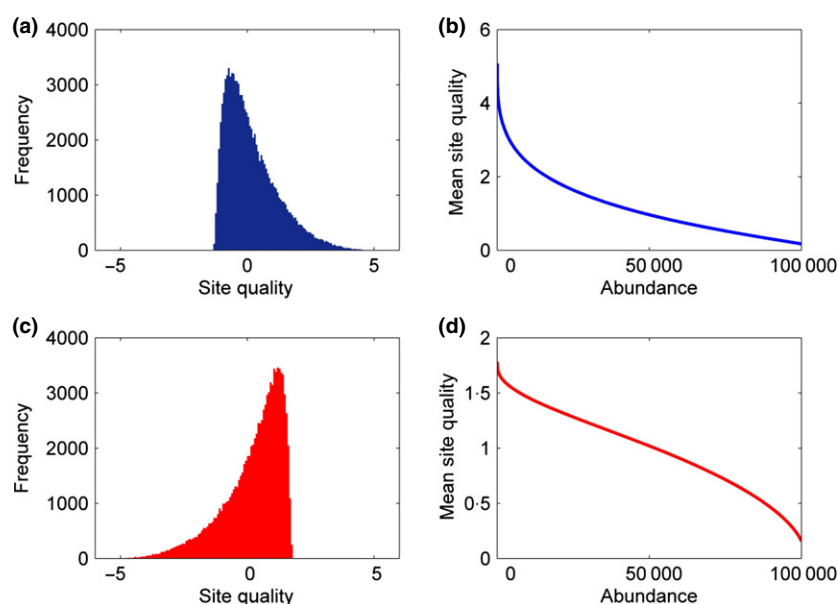


Fig. 7. Theoretical model of the effect of the statistical distribution of site qualities on population growth rate. Population growth rate is indexed by mean site quality. Individuals exclusively occupy the best available breeding site. A population with positive skewness in breeding site quality (a) will experience rapid decline in population growth rate (mean site quality) as abundance initially increases from a low level (b). A population with negative skewness in breeding site quality (c) will experience less decline in population growth rate as abundance increases from a low level (d) than a population with positive skewness (b). When a large population is reduced, population growth rate will increase more if there is negative skewness in habitat qualities than if there is positive skewness in habitat qualities.

valley-width habitats (Fig. 1c,f). Secondly, Chinook salmon may exhibit territorial behaviour with respect to stream gradient, but not valley width.

Additional work on density-dependent habitat selection across taxa is needed, so that evolutionary ecologists can better understand the conditions wherein an IFD is more likely than an IDD. Such work will be integral to the eco-evolutionary synthesis. For example, a population with an IDD should have different genetic dynamics than a population with an IFD. The IDD implies that dominant or early-arriving individuals produce disproportionately more offspring than subordinates. Thus, populations with IDD will have greater variance in family size and hence lower effective population size (Wright 1931, 1938) than a population with IFD. Furthermore, if recruitment from subordinates using marginal habitat is less than replacement, then source-sink dynamics (Pulliam 1988) emerge. When individuals born in a sink can reproduce in a source, niche evolution involves complex processes that are known to be incompletely understood (Holt 2011), but a general theoretical finding is that selection is less effective at producing individuals adapted to sink habitat when sinks habitats are more harsh (Holt 2011).

Population regulation can occur even when change in population size does not affect the survival or fecundity of individuals in particular breeding sites (Rodenhouse, Sherry & Holmes 1997). If individuals obey an IDD and use the best available breeding sites, then average per-capita fecundity will decline as population size increases (McPeck *et al.* 2001). A population occurring on a landscape with positive skewness in the statistical distribution of breeding site qualities will exhibit rapid initial declines in population growth rate as population size increases by a given amount from a low level (Fig 7, top row). However, when this population is large, a reduction in abundance by the same amount will produce relatively minor improvement in population growth rate (Fig. 7, top row). In contrast, a population on a landscape with negative skewness in the distribution of breeding site qualities will experience the opposite effects of change in abundance on population growth rate (Fig. 7, bottom row). The dynamics of population regulation under an IDD therefore depend not only on mean habitat quality, but also the skewness of the distribution of breeding site qualities.

Ecologists frequently associate local abundance to habitat features without regard to the potential effects of population size. Although it is difficult to collect the data needed to assess the effect of population size on density-habitat relationships, failure to do so can result in a snapshot in time of the density-habitat relationship that can be misleading in future applications. Although the density of animals is well known to be a potentially misleading indicator of habitat quality (Van Horne 1983; Kokko & Sutherland 2001; Batten 2004), the usefulness of density-habitat relationships to restoration planning is even further encumbered if dynamics with population size are unknown. Spawning habitat quality would be ideally

indexed by the number of offspring surviving to a maturity; however, this is difficult to measure. Assessing the curvature of isodars in different habitats attributes could reveal which habitat features are contested, which could be used to assess habitat quality. Here, greater curvature in the isodars of channel gradient than channel width could indicate that the former is a more important habitat attribute than the later.

U.S. Federal and state expenditures on Pacific salmon and steelhead recovery totalled \$1.1 billion from 2000 through 2009 (PCSRF 2011). Nonetheless, monitoring programmes designed to obtain unbiased and precise estimates of abundance, which are needed to assess conservation status and set appropriate harvest limits, are prohibitively expensive for many populations. Peak counts at select survey sites are therefore used as a surrogate index of abundance. The findings presented here suggest that stream flow obscures the relationship between peak counts and population size and that habitat selection behaviour causes peak counts to respond nonlinearly to changes in population size.

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Data accessibility

Data available from the Dryad Digital Repository: <http://dx.doi.org/10.5061/dryad.6g35k> (Falcý 2014).

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Supporting Information

Additional Supporting Information may be found in the online version of this article.

Fig. S1. Fourteen populations (blue polygons) on the coast of Oregon, USA (inset) where 2436 observations of spawning Chinook salmon density have been made at ca. 1.6 km units since 1986. Yellow diamonds indicate populations with a stream flow gauge. Arrows indicate which gauge was used to represent stream flow for populations that do not contain a stream flow gauge.